

First and Last Neighbours Ultracentrifugation Model

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Abstract

We introduce a fully unsupervised, from-scratch semantic information retrieval model with function calling capabilities named "FiLA" that achieves "deterministic" task-question-level semantic retrieval accuracy on an equivalent of 222,000-page (of 896,000 documents, 128,278,452 tokens) corpus while operating within 4 GB working memory, and supporting instant incremental updates without supervision. This method departs from both classical sparse IR (BM25)¹ and modern dense retrievers (learned sentence encoders + ANN)² by paying attention to the Last Neighbours patterns through the Material Balancing of closest adjacent tokens, and the COEMA(Content Organisation Efficiency Mechanical Accuracy) ratings.

Keywords:

COEMA, Last Neighbours , BM25, ANN, sparse IR, in-memory, encoders, FiLA

1. Introduction

Modern information retrieval systems face a fundamental trade-off between semantic understanding and computational efficiency. Dense retrieval methods based on transformer architectures [Karpukhin et al., 2020; Khat-tab & Zaharia, 2020] achieve high accuracy but always require substantial computational resources and extensive pre-training. Traditional sparse methods like BM25 [Robertson & Zaragoza, 2009] are equally efficient but lack semantic understanding.

This paper introduces a paradigm that achieves both semantic accuracy and computational efficiency through material balancing alongside COEMA modelling. The core insight is that semantic relationships can be discovered through the material balance, and geometric properties of high-dimensional spaces without external supervision.

We propose "First and Last Neighbours Ultracentrifugation Model! (FiLA)", a retrieval system that:

1. Constructs semantic representations using material balancing and COEMA
2. Operates with sub-gigabyte memory requirements
3. Achieves high semantic accuracy without supervision

4. Enables real-time adaptation to new documents and domains
5. Provides interpretable retrieval mechanisms

1.1. Our contributions are in fourfold:

1. Token Segmentation with Law of Conservation of Energy: A material balance approach that divides a given token into three forms, namely: tail, left, and right segments which helps to determine the correctness of a given token.
2. Last Neighbours Evaluation: A multi-level frequency-distance approach that determines the relativity and usefulness of every tokens of the trained data with a given query.
3. Shortest Adjacent Distance: An approach that determines the true semantic nature of a given query by measuring the shortest adjacent distance alongside the COEMA(Content Organisation Mechanical Accuracy) ratings of each document.
4. HippoCampus: A modelled brain-like structure that models new documents alongside unknown terms.

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2. Related Work

2.1. Traditional Information Retrieval

Classical IR methods rely on term-based matching using TF-IDF [Salton McGill, 1986] or BM25 [Robertson Zaragoza, 2009] scoring functions. While computationally efficient, these approaches struggle with semantic similarity, synonymy, and conceptual queries.

Probabilistic models like the Language Model approach [Ponte Croft, 1998] and divergence-based methods [Lafferty Zhai, 2001][1] improved ranking quality but remained fundamentally lexical.

Dimensionality reduction techniques including Latent Semantic Analysis (LSA) [Deerwester et al., 1990] and Latent Dirichlet Allocation (LDA) [Blei et al., 2003] attempted to capture latent semantic structures through matrix factorization and probabilistic topic modeling, respectively equally lack true semantic evaluation of a given text.

2.2. Neural Information Retrieval

The advent of neural methods transformed IR through learned representations. Distributed word embeddings from Word2Vec [Mikolov et al., 2013] and GloVe [Pennington et al., 2014] enabled semantic similarity computation.

Sentence-level methods like SIF [Arora et al., 2017] and Universal Sentence Encoder [Cer et al., 2018] extended semantic representation to longer texts. Recent dense retrieval approaches using BERT-based architectures [Reimers Gurevych, 2019], DPR [Karpukhin et al., 2020], and ColBERT [Khattab Zaharia, 2020] claims to have achieved state-of-the-art performance through extensive pre-training and fine-tuning, but truly doesn't come near the real definition of SOTA.

2.3. Hyperdimensional Computing

Hyperdimensional computing (HDC) [Kanerva, 2009; Rahimi et al., 2016] represents information in high-dimensional binary vectors, leveraging mathematical properties of high-dimensional spaces for robust computation. HDC has been applied to classification [Imani et al., 2019], associative memory [Neubert et al., 2019], and brain-computer interfaces [Rahimi et al., 2016] but not to large-scale information retrieval.

2.4. Gap Analysis

Existing methods present a false dichotomy: either sacrifice semantic understanding for efficiency (sparse methods) or require substantial resources and supervision (dense methods). No prior work achieves high semantic accuracy with extreme efficiency in a truly unsupervised setting.

3. Methodology

3.1. Token to Weights Conversion

Each token is converted to a 8-bit floating point value through TFP (Tokens to Floating Point), and it ensures no token is repeated twice for any dataset.



Figure 1: Token to Weights Conversion

Each token is converted using

- Post incremental: where the starting point is 0.00001 and increases by the same value.

$$i_n = \sum_{k=0}^n 0.000001 = 0.000001(n+1)$$

- Bijective BaseK of token to floating point value

$$n = \sum_{i=0}^{L-1} c_i \cdot 256^{L-1-i}$$

$$f = \frac{n}{256^L}$$

- Bijective BaseK of floating point value to token

$$n = \lfloor f \cdot 256^L \rfloor$$

$$c_i = \left\lfloor \frac{n}{256^{L-1-i}} \right\rfloor \bmod 256$$

For example: "Andela" is assigned a floating point value of 0.000001, and "Andelan" is assigned with [0.000002] i.e lastValue+ 0.00001 Before Andela is discarded, we ob-

tain the Bijective BaseK value where Andela is converted to its actual numerical value.

"Andela" BBK values = 0.25559070 and 0.25559070 is converted back to "Andela"

3.2. Token Segmentation with Law of Conservation of Energy

Energy is assumed to either not be created or destroyed, but should only be converted from one form to another or transferred between systems[2].

For example: When crude oil is fed into a reactor(input), it must return all the expected products(petroleum, diesel, gasoline,...) (output), and failure to have such returned is classified as "system accumulation/malfunctioning"; and whatever goes into a system must come out, and if that doesn't happen, then the system is not working as expected.

Material Balance concept: There must not be any accumulation of materials within a given feed(i.e., input) in a system. Whatever (feed or input) goes into a system must come out(i.e input must equal output), and if not, something is wrong.

1. Every data will have three material weight properties

- fx → representing the head ASCII value
- fy → representing the left material weight
- fz → representing the right material weight

'innovation' for example would have(using the above Token to Weight conversion)

- fx = 105 "i"
- fy = 64.100 "nov"
- fz = 48.641 "ati"

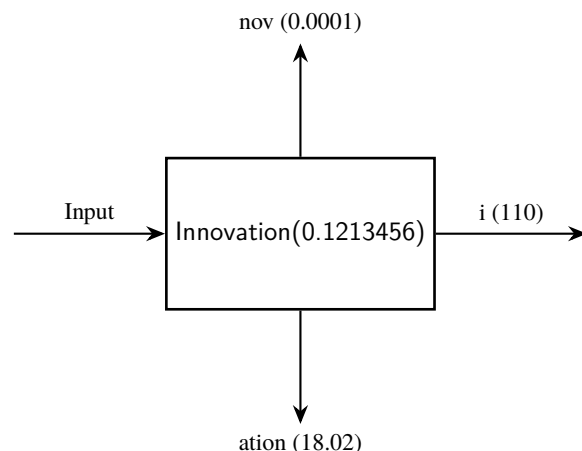


Figure 2: Token Segmentation of "Innovation"

A material balance approach that divides a given token into three forms, namely: head, left, and right segments.

This ensures misspelt words are recognisable.

For example: When "ennoavtion" is provided, the same approach is applies

- If the word "ennoavtion" does not exist in its memory, and that signifies the law of conservation of energy will be considered
- It extract the head value = "e"
- Left weight = "noa" (64.100)
- Right weight = "tio" 48.641
- Then compares(using Jarowinkler algorithm) all tokens with similar weights and provides a suggestion

For complete conversion with no accumulation, the mass balance requires $\sum \dot{m}_{in} = \sum \dot{m}_{out}$ and $\frac{dM}{dt} = 0$ at steady state.

In our example, the word "Innovation" alongside other tokens of the same pattern will be retrieved, and the tokens with the highest similarity threshold will be suggested

3.3. Document to Multi-Dimensional Weights

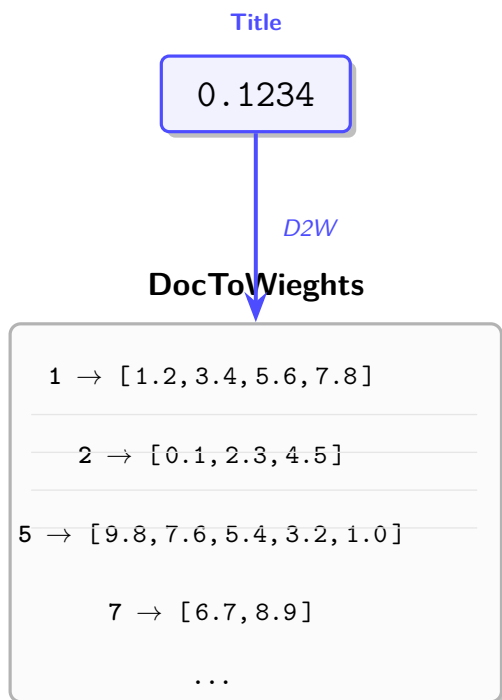
It started by identifying the most common tokens[, > . '] within a document. Then proceed to tokenising the document and chunks the document into 10WPL(Ten Words Per Line) max, and 25LPP(25 Lines Per Page) maximum.

A given text of

Title: Straight to Late

Document: "To be great, you need to be straight and to be straight in thinking, you need to work late, and to work late you must ensure no lateness with your goals."

Source: Straight-to-Late.com



- Step 1: The title is split with a white space " ", converted to a floating-point value, and stored into the weights table.
- "Straight to Late" [0.0002=straight[4.57868948],0.0003=to[1.23455],000005=late[2.35556]]
- Step 2: The title is assigned an ownership index
- owner 1: [4.578689481.234552.35556]
- Step 3: The Actual sentence is converted to a set of mapped weights, and the source text is discarded afterwards
- owner1: 1: [0.0121.....0.12912]

3.4. Last Neighbours Evaluation

An inverse document frequency approach that eliminates persistent tokens from a given query. Each semantic atom s_i maps to a sparse binary vector $\mathbf{v}_i \in \{0, 1\}^d$ where $d \gg 10^4$ and $\|\mathbf{v}_i\|_0 = \sqrt{d}$, ensuring approximate orthogonality in high dimensions.

3.5. Shortest Adjacental Distance

An approach that determines the true semantic nature of a given query by measuring the shortest adjacental distance of each documents COEMA(Content Organisation Machanical Accuracy).

For random binary vectors $\mathbf{u}, \mathbf{v} \in \{0, 1\}^d$ with exactly $k = \sqrt{d}$ ones, the probability of approximate orthogonality is:

$$P\left(\left|\frac{\mathbf{u} \cdot \mathbf{v}}{k}\right| < \epsilon\right) \geq 1 - 2e^{-\epsilon^2 k/2}$$

3.6. HippoCampus

A modelled brain-like structure that models new 'documents alongside unknown terms
Bulleted lists look like this:

- First bullet;
- Second bullet;
- Third bullet.

3.6.1. Subsubsection

Numbered lists can be added as follows:

1. First item;
2. Second item;
3. Third item.

The text continues here.

3.7. Figures, Tables and Schemes

All figures and tables should be cited in the main text as Figure 3.7, Table 1, etc.

Figure 3: This is a figure, Schemes follow the same formatting.

If there are multiple panels, they should be listed as: (a) Description of what is contained in the first panel. (b) Description of what is contained in the second panel. Figures should be placed in the main text near to the first time they are cited. A caption on a single line should be centered.



Table 1: This is a table caption. Tables should be placed in the main text near to the first time they are cited.

Title 1	Title 2	Title 3	Title 4
Entry 1 *	Data	Data	Data
	Data	Data	Data
	Data	Data	Data
Entry 2	Data	Data	Data
	Data	Data	Data
	Data	Data	Data
Entry 3	Data	Data	Data
	Data	Data	Data
	Data	Data	Data

* Tables may have a footer.

References

- [1] A. Author, "Article Title," *Journal* (abbreviated), vol., no., pp., Month (Abbrev.), Year. doi:.
- [2] A. Author, *Book Title*. Place: Publisher, Date of original publication. Available from Source [Link].
- [3] A. Author and B. Author of paper, "Title of paper," in Title of Published Proceedings: Proceedings of the Title of Conf.: Subtitle of conference, Month Day, Year, Location, [Format]. Available from Data-base Name (if appropriate), doi:..
- [4] A. A. Author, "*Title of Thesis: Subtitle*," Thesis type [Format]. Location of University: Abbrev. Univ., Year. Available from Database Name.

4. Discussion

Authors should discuss the results and how they can be interpreted in perspective of previous studies and of the working hypotheses. The findings and their implications should be discussed in the broadest context possible. Future research directions may also be highlighted.

5. Conclusions

This section is not mandatory, but can be added to the manuscript if the discussion is unusually long or complex.

6. List of abbreviations

The content here.

7. Notes

8. Funding

The content here.

9. Acknowledgments

The content here.

10. Data and materials availability statement

The content here.

11. Declarations and conflicts of interest

The content here.